MICHIGAN
INTERNATIONAL
PREP SCHOOL

2026–2027 ACADEMIC YEAR

School of Technology Course Catalog

Prepare for the future through hands-on technology courses that build real world skills in programming, cybersecurity, information technology, web development, and artificial intelligence. Whether you're exploring technology for the first time or preparing for an industry certification, there's a pathway designed for you.

Every course is **one semester long** and **asynchronous**, allowing students to work at a pace that fits their schedule. To stay connected, students are encouraged to join **optional weekly video call sessions** where they can collaborate with classmates, showcase projects, get help from their instructor, and participate in discussions about emerging technology.

Each course listing includes its semester availability.

S1 - offered only during Fall semester.
S2 - offered only during Winter semester.
S1 or S2 - offered during either semester.

QUESTIONS? GET IN TOUCH

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TRACK

Foundations of Programming

Formerly Exploring Tech I

Exploring Technology and Art

S1 GRADES 6–12

Combine creativity with computer science to build your own interactive games and music. You'll begin using Code.org's Game Lab environment to program shapes, sprites, and animations, handle user input, and use variables and conditionals to turn simple images into complex, multi-level games.

Then you'll explore foundational programming concepts through music, using code to build, remix, and perform original songs. Through sequencing, loops, and functions, you'll practice computational thinking like pattern recognition, algorithmic thinking, and debugging to design, test, and refine your musical ideas.

Formerly Exploring Tech II

Intro to Python

S2 GRADES 6–12

An introduction to the fundamentals of computer science through visual programming and text-based logic. In CodeHS you'll master top-down design, functions, and decomposition to solve grid-based puzzles and build a strong foundation in algorithmic thinking and problem-solving.

As you transition to standard Python, you'll manage data with variables, user input, and mathematical operators, then master control structures including if/else statements and logical operators. The course concludes with looping and nested logic, where you'll build interactive projects like a Mad Libs generator, a Quiz Game, and a Password Authenticator.

TRACK

Cybersecurity

Intro to Cybersecurity

S1 OR S2 GRADES 7–12

Build, manage, and protect a secure business network. You'll master identity management, controlling user access on Windows and Linux and enforcing strong password policies, then get hands-on with network hardening by installing firewalls and setting up VPNs for secure remote connections.

Beyond hardware, you'll defend computers with antivirus software, application whitelisting, and file permissions, and protect data through encryption and critical backups. To keep systems safe over time, you'll run vulnerability scans, monitor system logs, and analyze network attacks to stop threats before they cause damage.

Ethical Hacker

S1 OR S2 GRADES 7–12

Learn the professional process used to test and secure modern digital networks. Starting with the "Prepare" and "Reconnaissance" phases, you'll use operating-system tools and specialized software to find devices on a network, including wireless and IoT hardware, then perform vulnerability scans and enumeration to find open ports and active users.

As you progress, you'll practice gaining access by cracking passwords, using social engineering, and exploiting web vulnerabilities like SQL injection. You'll pick up post-exploitation skills such as privilege escalation and session hijacking, and study how attackers cover their tracks, while learning to defend systems with honeypots and intrusion detection systems (IDS).

CyberDefense Pro

S1 OR S2 | GRADES 9–12

PREREQUISITE Pass Introduction to Cybersecurity or Ethical Hacker first.

An advanced course that prepares you for a career as a cybersecurity analyst. You'll detect and respond to modern threats by mastering vulnerability management, threat hunting, and network security, using professional tools like SIEMs and protocol analyzers to monitor traffic, identify indicators of compromise, and defend against attacks like SQL injection and ransomware.

With an emphasis on incident response and digital forensics, you'll learn to contain breaches and analyze evidence, and explore specialized security for cloud, wireless, and IoT systems, all while preparing for the CompTIA CySA+ (CS0-003) and TestOut CyberDefense Pro certification exams.

TRACK

IT & Hardware

IT Fundamentals

S1 OR S2 | GRADES 7–12

A broad introduction to the modern world of technology, preparing you for the CompTIA Tech+ certification. You'll start with computing basics, data privacy, and internet technologies, then cover the essential networking and cybersecurity concepts that keep data moving safely across the globe.

From there you'll dive into hardware, software, and databases to understand how computers are built and store information, get a head start in coding and logic, and look ahead at emerging technologies including AI, robotics, and virtual reality.

PC Pro A

S1 GRADES 9–12

A comprehensive introduction to IT that prepares you for industry certifications like TestOut PC Pro and CompTIA A+. You'll learn the core responsibilities of a PC technician, using professional ticketing systems, managing documentation, and following safety and maintenance procedures.

The focus is hands-on hardware: installing, configuring, and troubleshooting motherboards, processors, and power supplies. You'll explore Windows, Linux, and macOS, manage storage systems like RAID and partitioning, and master system implementation by performing full Windows installations and building virtual machines.

PC Pro B

S2 GRADES 9–12

PREREQUISITE Pass PC Pro A first.

The advanced skills needed to manage and protect a professional IT environment. You'll master Windows and Linux management through system tools, scripting, and the command line, including managing users in Active Directory, setting up VPNs for remote work, and performing critical data backups.

You'll also explore networking and mobile support, configuring secure Wi-Fi and troubleshooting devices from laptops to 3D printers, then finish with a focus on cybersecurity: configuring firewalls, blocking malware, and defending against social engineering to keep an organization's data safe.

TRACK

Web & Emerging Tech

Web Design and Dev I S1 GRADES 7–12

Learn the foundational skills to create, style, and publish your own professional websites. You'll explore how web pages are developed and delivered across the globe, then start building right away with HTML, mastering tags, attributes, and elements to structure pages with images, hyperlinks, and data tables.

Next you'll style your creations with CSS, using selectors and classes to manage layout, color, and fonts and understanding how the "cascade" sets visual priority. You'll design a live homepage portfolio from scratch, then finish with advanced techniques: divs and spans for precise layout, embedded media through iframes, and CSS animations that make sites dynamic.

Web Design and Dev II S2 GRADES 7–12

PREREQUISITE Pass Web Design and Dev I first.

Transition from basic page construction to building professional, responsive, user-centered experiences. You'll start with the Bootstrap framework, using open-source components and grid systems to create mobile-responsive layouts that adapt to any screen, while practicing how to read and implement real technical documentation.

The focus then shifts to UI and UX design: researching stakeholder needs, defining clear problem statements, rapid prototyping from paper sketches to digital solutions, and user testing with peers. The course concludes with a comprehensive final project where you design, prototype, and develop a website from scratch.

Artificial Intelligence

S1 OR S2

GRADES 9–12

A hands-on exploration of how AI works, from basic logic to building your own neural networks. You'll compare human and artificial intelligence, test the limits of Large Language Models, and master Prompt Engineering to generate high-quality text and images, while examining how companies build AI and the critical issues of bias and ethics.

The second half dives into Machine Learning through interactive projects: using tools like TensorFlow and Google Colab, you'll build and train models to solve math problems, classify images (cats vs. dogs), and perform Natural Language Processing on movie and hotel reviews. The course closes on the risks and future of AI, identifying deepfakes, defending against prompt-injection attacks, and debating AI's legal and social impact in a mock trial.

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